# E350

# U1300 and U3400

RELEASE NOTES FOR F/W VERSION R5.0

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Filename: E350 U1300 and U3400 Release Notes
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Revision history

| Version | Date | Comments |
|---|---|---|
| 1 | 1/08/2018 | Initial Release |
| 2 | 21/08/2018 | Updated to include, tamper detection bug (TFS-11024) |
| 3 | 29/08/2018 | Minor update |

**Important Information**
This document provides information relating to E350 U1300 Series 1, Series 2 and U3400 meter firmware (the "Product")

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# 1 About this document

## 1.1 Purpose & Scope

The purpose of this ***R5.0 Firmware Release Note*** is to communicate new features, limitations and changes in meter functionality.

The R5 firmware release is for the E350 - U1300 S1, U1300 S2 and U3400 meters.

## 1.2 Definitions, Acronyms, and Abbreviations

| | |
|---|---|
| RSM: | Relay State Machine for the load control relay |
| LC: | Load Control |
| LP: | Load Profile |
| CLC/R: | Controlled Load Contactor or Relay |
| CSM: | Customer Supply Monitoring including Neutral Integrity |
| QoS: | Quality of Supply |
| AMI: | Advanced Metering Infrastructure |
| AD: | Auto Disconnect |
| S2: | Series 2 |
| SCC: | Supply Capacity Control |
| STS: | Smart Time Synchronisation |
| HES: | Head End Systems |
| E350: | Family name of Landis+Gyr modular meter including the types U1200, U1300, U1300 S2 U3300, U3400 and U3350 |
| E650: | Family name of Landis+Gyr commercial and industrial meter including the type EM5400 and EM5400 S2 |
| E420: | Family name of Landis+Gyr integrated meter S1100 RF and S1100 LTE |
| R1100: | Landis+Gyr’s smart load control device. |

## 1.3 References

E350 U1300 S1 and U1300 S2 User Manual

E350 U3400 User Manual

EMPWin+ User Manual

# 2 About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released firmware version: R4.9

This Released Version: R5.0

| Compatible Meter | Firmware Upgrade* | Firmware Downgrade* |
|---|---|---|
| U1300 Series 1 | 4.2 → 5.0<br>↓<br>4.3 → 5.0<br>↓<br>4.4 → 5.0<br>↓<br>4.5 → 5.0<br>↓<br>4.6 → 5.0<br>↓<br>4.8 → 5.0<br>↓<br>4.9 → 5.0 | Not supported. |
| U1300 Series 2 | 4.4 → 5.0<br>↓<br>4.5 → 5.0<br>↓<br>4.6 → 5.0<br>↓<br>4.8 → 5.0<br>↓<br>4.9 → 5.0 | Not supported. |
| U3400 | 4.4 → 5.0<br>↓<br>4.5 → 5.0<br>↓<br>4.6 → 5.0<br>↓<br>4.7 → 5.0<br>↓<br>4.9 → 5.0 | Not supported. |

**Important Note:** *customer will be required to refer to the release notes for the particular firmware version to understand the limitations of the upgrade/downgrade. Please also refer to the intermediary firmware release notes when skipping meter firmware version.

This firmware has been tested on the following platforms:

- E350 U1300 meters (this includes Series 1 and Series 2)
- E350 U3400 meters
- EMPWin+ 3.3.3

The FT100 release 5.1.2 or earlier must not be used with R5.0 meter firmware.

EMPWin+ 3.3.3 or greater must be used with R5.0 meter firmware.


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## 2.1 New Features for Supported Meters

### 2.1.1 General New Features

- The E350 U1300 & U3400 with R5.0 firmware now supports:
  - Smart Time Synchronisation (STS) backwards by small increments. Enhanced event reporting has been implemented for time change information.
  - Self- read periods in TOU to have 35 snapshots.
  - 5-minute load profiling interval selection.
  - The repartitioning of unused memory to extend the Load Profile storage capacity. The “Extended 1” memory size increase the storage to support 5 minute Load profile data. The “Extended 1” storage size allow up to 220 days of 5 minute LP for a single LP group with four channels.
  - “Extended 1” memory size and 5 minute intervals can be selected via the EMPWin+ MeterDev. The original LP memory size in meters with firmware R4.9 or earlier is known as “Standard Memory” size.

### 2.1.2 Firmware Upgrade

- The E350 U1300 & U3400 meters can be F/W upgraded from any R4.x to R5.0.
- The LP data is maintained when the firmware upgrade process expands the available LP memory. However, a reprogram is required to utilise this memory.
- The TOU data is maintained when the firmware upgrade process expands the TOU area from 8 to 35 days of self-reads. Previous Billing data can be corrupted when F/W is upgraded from any R4.x to R5.0. In most applications the Previous Billing data is not used by HES systems or by customers. Re-programming the meter is required to enable the new functionality provided in R5.0 and will also clear the Previous Billing data.

### 2.1.3 Firmware Downgrade

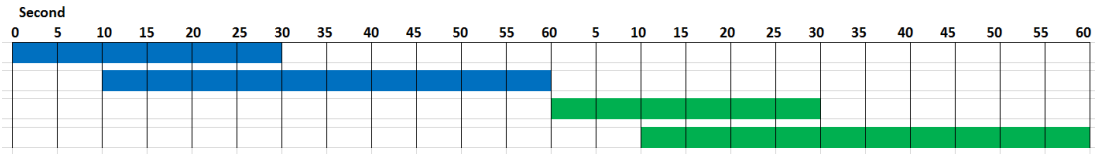
Downgrade from R5.0 to R4.x is not supported due to a hardware flag being automatically set when the meter is upgraded to R5.0. This cannot be changed by the customer when downgrading the F/W back to R4.x. If a downgrade is attempted, Customers will notice a program read can’t be performed (with EMPWin+ version 3.3.3 or greater) and if an earlier EMPWin+ version were used, they will see the TOU and LP data will become corrupted. This is due to the structural changes required in order to support 35 TOU self-read periods and additional LP area that was created during the R5.0 F/W upgrade.

### 2.1.4 Billing Self Read Periods

- E350 U1300 & U3400 meters running R5.0 F/W will record 35 self-read periods.

### 2.1.5 Clock Calendar

- E350 U1300 & U3400 meters with R5.0 F/W supports the new Smart Time Synchronisation (**STS**) functionality. In all previous F/W versions a time change back across a completed LP interval boundary could occur and LP recording would be suspended.
    - The **STS** functionality makes small time changes of up to 20 seconds backwards every 60 seconds. Therefore, the meter will take up to 80 seconds for every minute recorded by the meter until it has completed its time update.




    - The **STS** method is not used for forward time changes.
    - The **STS** method is not used if the requested time change backwards and is greater than one hour
    - The **STS** method is not used if the backwards time change can be performed with in the interval boundary.

- If the **STS** functionality has started to make the small incremental changes and then a power outage occurs the process will be terminated and no “Time Change To” event will be created. The outage will leave the meters’ time partially adjusted so another time change request may need to be performed.
- If **STS** functionality is started, the required change in mm:ss are logged in the event log.

**Events**

| Time | Description |
|---|---|
| 11/05/2018 13:01:28 | Time Changed (new time used as time stamp) |
| 11/05/2018 12:55:19 | Time Changed (old time used as time stamp)(Slow time adjustment mode backwards by 02:02) |

Note: The STS time adjustment will typically take several minutes to complete, and the event times will look like a forward time change occurred. Please refer to the event log description for indication of STS event and required time change in mm:ss.

### 2.1.6 Load Profile

- The E350 U1300 & U3400 with R5.0 firmware now supports 5 minute Load profile interval selection and increased memory referred to as “Extended 1”. The extended memory can store up to 220 days of 5 minute LP intervals for a single LP group with four channels enabled.
- Meter configuration (MCD) files with 5 minute intervals enabled are restricted in MeterComm and MCDExport from being written to meters with older firmware.

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## 2.2 Command Center

Command Center (CC) version 7.2.0.8 has been tested with R5.0 F/W

- The meter F/W can be sent Over the Air (OTA) using CC.
- R4.9 and R4.8 to R5.0 was tested using RF comms 14.01.001
- U1300 OTA R4.8 to R5.0 Cellular comms 14.12.001 was OK.
- Meter will continue to push LP, Billing Reads and Events to CC.
- CC can perform a complete read of the 35 self-read periods
- CC can read LP data from meters that have 5, 10, 15, 30 or 60 minute LP intervals.
- The new Smart Time Synchronisation (**STS**) method in R5.0 F/W is compatible with CC methods of adjusting time.

## 2.3 Bug fixes in this release

| *Item No.* | *Description* |
|---|---|
| **TFS-1036** | When input voltage is below the minimum operating voltage for the whole of the LP interval length then the minimum LP Voltage value could be reported as an overflow value because it was never updated, this is now fixed and is correctly reported as 0. |

## 2.4 Known Issues in this release

| *Item No.* | *Description* |
|---|---|
| **Mantis-3163** | If there is a long power outage across a Season change date, then the Billing report might incorrectly label the season number as 0. The date logged, and energy/demand values are correct. This is an existing bug from previous firmware versions. |
| **Mantis-3584** | If there is a power outage across a LC season change from LC season 1 to LC season 0 then the event that indicates this LC season change to 0 is not recorded. The logging works correctly for LC season change from 0 to 1 across a power outage. The LC scheduler works correctly, it is only the event that is missing. |
| **Mantis-3096** | <u>Demand Limit Load Control</u><br><br>If Under Frequency Load Control (UFLC) module is triggered due to bad frequency, then the frequency recovers and UFLC starts its Nominal Delay period AND then a short power cycle occurs (about 3 seconds) AND on Power Up the Frequency is still within Normal limits; under these conditions the UFLC module might not correctly exit its Nominal waiting period. This can cause the UFLC module to block other lower priority RSM commands such as Ripple telegrams. It appears that when this incorrect state is reached that it might require two power cycles to correctly reset the RSM logic. |
| **TFS-6268** | If the meter U1300 and U3400 meter with R5.0 firmware is running in SSN mode and a backwards time reset over the interval boundary is actioned, only the start time event is recorded in the log. |

| | |
|---|---|
| **TFS-2519** | <u>Corruption of Previous Billing data when firmware upgrading to R5.0</u><br><br>Meters can be F/W upgraded from any R4.x to R5.0. The TOU data and LP data is maintained when the firmware upgrade process expands the TOU area from 8 to 35 days of self-reads. Previous Billing data can be corrupted when F/W upgraded from any R4.x to R5.0. In most applications the Previous Billing data is not used by HES systems or by customers. Re-programming the meter is required to enable the new functionality provided in R5.0 and will also clear the Previous Billing data. |
| **TFS-1565**<br>**Mantis-3918** | It has been observed that U1300 meters, the LC1 & LC2 random delay closed values tend to have the same minutes and seconds with only the hours being different. Random delay is not random enough for the two relays. |
| **TFS-1067** | It is possible that after installation the meter might not switch the LC relay when the first Ripple Telegram is received. If the meter has been programmed with Ripple, then a power outage occurs before the first valid telegram is sent, then the LC relay might not switch. Subsequent ripple telegrams are handled correctly.<br>After the first telegram the meter works correctly. Utilities usually send multiple telegrams even of the same type throughout the day, so this should not have an impact on the consumer. |
| **TFS-1567**<br>**Mantis-3904** | If the Ripple Power Restore option has been set to Time Table AND the random power restore range has been set to 0 then the Time Table schedule will not run. This is unlikely to occur because the random power restore is not likely to be set to 0 by customers. |
| **TFS-11024** | A bug has been found on R5.0 that tamper functionality has been disabled for U3400 meter. This is planned to be fixed in R5.4 F/W. |


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